

# Hierarchy of infinities



# Outline

Comparison of infinities for functions



# Comparison of infinities for functions



# Introduction: comparison of infinities for functions

- ▶ In this section we compare the **speed** at which three families of functions tend to  $+\infty$ : **logarithms**, **powers** and **exponentials**
- ▶ The result is a true **hierarchy**:  $\text{logarithms} \ll \text{powers} \ll \text{exponentials}$
- ▶ With a change of variable we will obtain many useful limits also as  $x \rightarrow 0^+$  and as  $x \rightarrow -\infty$ , which allow us to solve indeterminate forms of the type  $0 \cdot \infty$  and  $\frac{\infty}{\infty}$

# Comparison of infinities

## Theorem 1: Comparison of infinities

Given  $\alpha, \beta, \lambda > 0$ ,  $a, b > 1$ , we have

$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

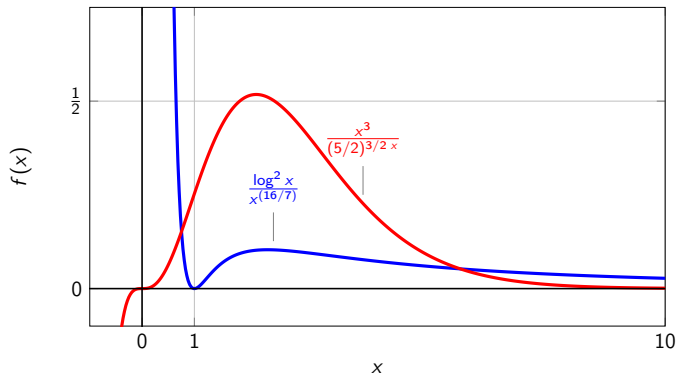
**Proof:** the theorem will be proved with the tools of differential calculus.

# Hierarchy of infinities

- ▶ Any power with positive exponent is an infinity of higher order than any power of logarithms with base  $> 1$ , and
- ▶ Any exponential with base  $> 1$  is an infinity of higher order than any power with positive exponent.

## Example

$$\lim_{x \rightarrow +\infty} \frac{\log^2 x}{x^{(16/7)}} = 0, \quad \lim_{x \rightarrow +\infty} \frac{x^3}{(5/2)^{3/2} x} = 0$$



## Comparison of infinities with a function $\eta(x)$

If a function  $\eta(x) \rightarrow +\infty$  as  $x \rightarrow c \in \mathbb{R}^*$ , we have, from the comparison of infinities theorem:

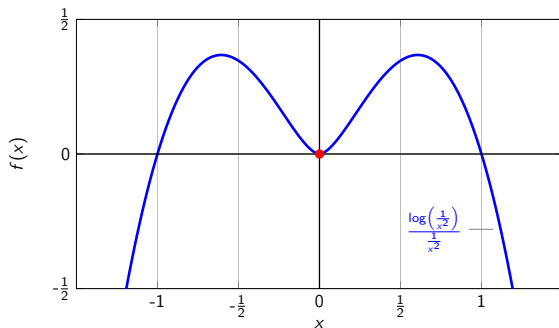
$$\lim_{x \rightarrow c} \frac{\left( \log_a (\eta(x)) \right)^\beta}{(\eta(x))^\alpha} = 0 \quad \text{and} \quad \lim_{x \rightarrow c} \frac{(\eta(x))^\alpha}{b^\lambda \eta(x)} = 0$$

with  $\alpha, \beta, \lambda > 0, a, b > 1$

## Example

$$\lim_{x \rightarrow 0} \frac{\log\left(\frac{1}{x^2}\right)}{\frac{1}{x^2}} = 0$$

since, with  $\eta(x) = \frac{1}{x^2}$ , we have  $\eta(x) \rightarrow +\infty$  as  $x \rightarrow 0$ .



With the change of variable  $t = \frac{1}{x}$  we now turn limits as  $x \rightarrow +\infty$  into limits as  $t \rightarrow 0^+$ : in this way we obtain a series of consequences of the theorem.

# Powers and logarithms at $0^+$

## Observation 1

$$\lim_{x \rightarrow 0^+} x^\alpha \log_a^\beta x = 0$$

$$\left( \alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1 \right)$$

# Proof

Given  $\alpha, \beta > 0, a > 1$  we have:

$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = 0 \quad \text{and} \quad \frac{\log_a^\beta x}{x^\alpha} = \left(\frac{1}{x}\right)^\alpha \left(-\log_a \left(\frac{1}{x}\right)\right)^\beta$$

Setting  $t = \frac{1}{x}$ ,  $x \rightarrow +\infty$  is equivalent to  $t \rightarrow 0^+$ , and substituting we have:

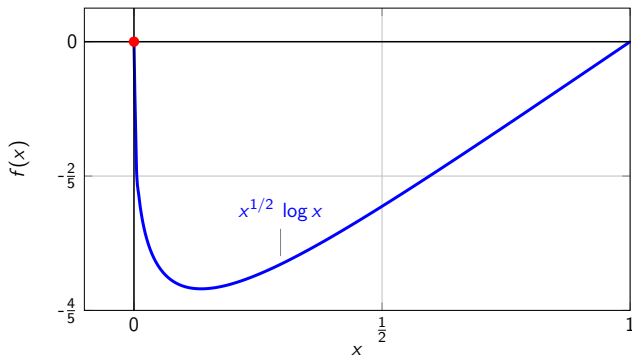
$$\lim_{x \rightarrow +\infty} \frac{\log_a^\beta x}{x^\alpha} = (-1)^\beta \lim_{t \rightarrow 0^+} t^\alpha \log_a^\beta t = 0$$



## Example (1/2)

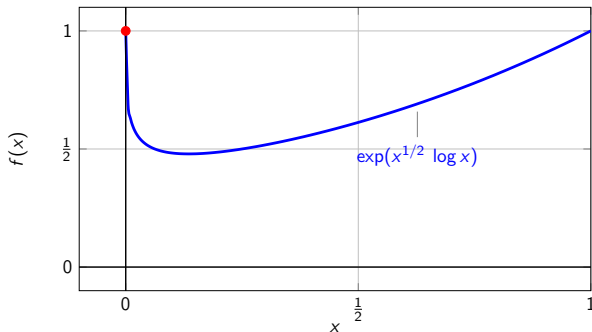
$$\lim_{x \rightarrow 0^+} x^{1/2} \log x = 0^-$$

since we have  $\alpha = \frac{1}{2} > 0$ ,  $\beta = 1 > 0$  and  $a = e > 1$ .



## Example (2/2)

$$\lim_{x \rightarrow 0^+} x^{\sqrt{x}} = 1^- \quad \text{since} \quad \lim_{x \rightarrow 0^+} x^{\sqrt{x}} = \lim_{x \rightarrow 0^+} \underbrace{e^{x^{1/2} \log x}}_{=\exp(x^{1/2} \log x)} = e^{0^-} = 1^-$$



## Powers and logarithms with a function $\varepsilon(x)$

If a function  $\varepsilon(x)$  tends to  $0^+$  as  $x \rightarrow c \in \mathbb{R}^*$ , it follows from the previous remark that:

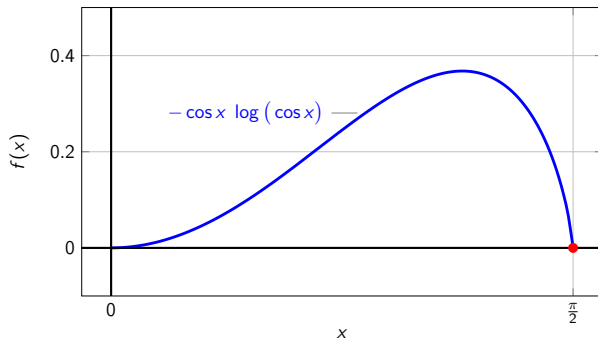
$$\lim_{x \rightarrow c} (\varepsilon(x))^\alpha (\log_a (\varepsilon(x)))^\beta = 0$$

$$\left( \alpha > 0, \beta = \frac{m}{n}, n, m \in \mathbb{N}, n \text{ odd}, a > 1 \right)$$

## Example

$$\lim_{x \rightarrow \frac{\pi}{2}^-} -\cos x \log(\cos x) = 0^+$$

since, with  $\varepsilon(x) = \cos x$ , we have  $\varepsilon(x) \rightarrow 0^+$  as  $x \rightarrow \frac{\pi}{2}^-$  and  $\alpha = \beta = 1$ .



# Limits of the reciprocals

From the comparison of infinities theorem it follows that:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{\log_a^\beta x} = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty$$

$$\text{with } \alpha, \beta, \lambda > 0, \quad a, b > 1$$

## Powers and exponentials at $0^+$ (1/2)

### Observation 2

$$\lim_{x \rightarrow 0^+} x^\alpha b^{\lambda/x} = +\infty \quad (\alpha, \lambda > 0, b > 1)$$

# Proof

Given  $\alpha, \lambda > 0, b > 1$ , we have:

$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = +\infty$$

Setting  $t = \frac{1}{x}$ ,  $x \rightarrow +\infty$  is equivalent to  $t \rightarrow 0^+$ , and substituting we have:

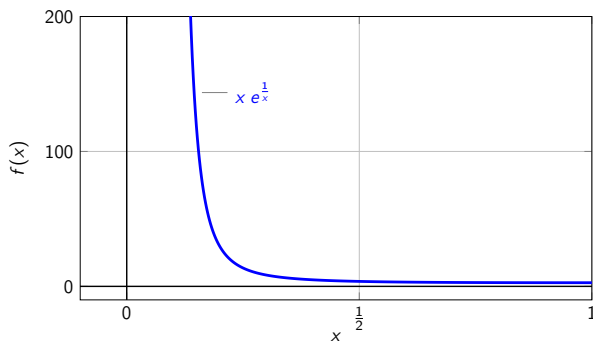
$$\lim_{x \rightarrow +\infty} \frac{b^{\lambda x}}{x^\alpha} = \lim_{x \rightarrow +\infty} \left(\frac{1}{x}\right)^\alpha b^{\lambda x} = \lim_{t \rightarrow 0^+} t^\alpha b^{\lambda/t} = +\infty$$



## Example

$$\lim_{x \rightarrow 0^+} x e^{\frac{1}{x}} = +\infty$$

since we have  $\alpha = \lambda = 1$  and  $b = e > 1$



## Powers and exponentials at $0^+$ (2/2)

### Observation 3

$$\lim_{x \rightarrow 0^+} x^{-\alpha} b^{-\lambda/x} = 0 \quad (\alpha, \lambda > 0, b > 1)$$

# Proof

Given  $\alpha, \lambda > 0, b > 1$  we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting  $t = \frac{1}{x}$ ,  $x \rightarrow +\infty$  is equivalent to  $t \rightarrow 0^+$ , and substituting we have:

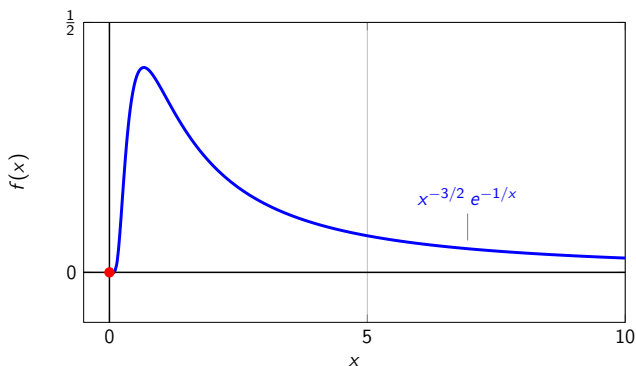
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{x \rightarrow +\infty} \left( \frac{1}{x} \right)^{-\alpha} \frac{1}{b^{\lambda x}} = \lim_{t \rightarrow 0^+} t^{-\alpha} b^{-\lambda/t} = 0$$



## Example

$$\lim_{x \rightarrow 0^+} x^{-3/2} e^{-1/x} = 0$$

since we have  $\alpha = \frac{3}{2}$ ,  $\lambda = 1$  and  $b = e > 1$ .



Finally, with the change of variable  $t = -x$  we turn limits as  $x \rightarrow +\infty$  into limits as  $t \rightarrow -\infty$ .

# Powers and exponentials at $-\infty$

## Observation 4

$$\lim_{x \rightarrow -\infty} x^\alpha b^{\lambda x} = 0$$

$$\left( \alpha = \frac{m}{n} > 0, n, m \in \mathbb{N}, n \text{ odd}, m \neq 0, \lambda > 0, b > 1 \right)$$

# Proof

Given  $\alpha, \lambda > 0, b > 1$  we have:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = 0$$

Setting  $t = -x$ ,  $x \rightarrow +\infty$  is equivalent to  $t \rightarrow -\infty$ , and substituting we have:

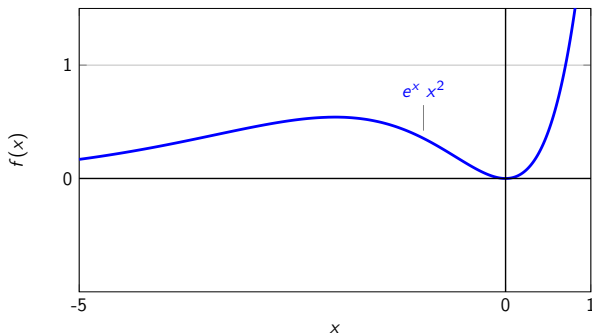
$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{b^{\lambda x}} = \lim_{t \rightarrow -\infty} \frac{(-t)^\alpha}{b^{-\lambda t}} = (-1)^\alpha \lim_{t \rightarrow -\infty} t^\alpha b^{\lambda t} = 0$$



## Example

$$\lim_{x \rightarrow -\infty} x^2 e^x = 0$$

since we have  $\alpha = 2$ ,  $\lambda = 1$  and  $b = e > 1$ .



# Summary (1/2)

**Hierarchy of infinities** (as  $x \rightarrow +\infty$ , with  $\alpha, \beta, \lambda > 0$ ,  $a, b > 1$ ):

- ▶  $\log_a^\beta x \ll x^\alpha \ll b^{\lambda x}$ : logarithms grow more slowly than powers, powers more slowly than exponentials
- ▶ The same limits hold replacing  $x$  with a function  $\eta(x) \rightarrow +\infty$
- ▶ The reciprocals tend to  $+\infty$

**Powers and logarithms at  $0^+$ :**

- ▶  $x^\alpha \log_a^\beta x \rightarrow 0$  as  $x \rightarrow 0^+$  (also with  $\varepsilon(x) \rightarrow 0^+$  in place of  $x$ )

## Summary (2/2)

### Powers and exponentials:

- ▶  $x^\alpha b^{\lambda/x} \rightarrow +\infty$  and  $x^{-\alpha} b^{-\lambda/x} \rightarrow 0$  as  $x \rightarrow 0^+$
- ▶  $x^\alpha b^{\lambda x} \rightarrow 0$  as  $x \rightarrow -\infty$

### Method:

- ▶ All these consequences are obtained from the comparison of infinities theorem with the changes of variable  $t = \frac{1}{x}$  and  $t = -x$
- ▶ They allow us to solve indeterminate forms of the type  $0 \cdot \infty$  and  $\frac{\infty}{\infty}$